PARTH MALIK

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EDUCATION

NORTHEASTERN UNIVERSITY, Boston, MA — Sep 2022 – Dec 2024
Master of Science in Data Science — *GPA: 3.92 / 4.0*

DEAKIN UNIVERSITY, Melbourne, Australia — Jul 2016 – Nov 2019
Bachelor of Commerce in Finance and Accounting — *GPA: 3.83 / 4.0*

TECHNICAL SKILLS

Languages: Python (NumPy, SciPy, TensorFlow, Keras, Scikit-Learn, PySpark), SQL, R(ggplot2, dplyr), MATLAB, VBScript
Tech Tools: Tableau, MLflow, Kubeflow, DVC, Airflow, Apache Spark, Hadoop, Kafka, MS Excel, Google Analytics
Databases: MySQL, MongoDB, Redis, Snowflake DB, MS-SQL Server, Oracle DB

EXPERIENCE

INTERNATIONAL DATA CORPORATION (IDC), Needham, MA
Data & Analytics Intern — Jun 2024 – Aug 2024

- Transformed private company revenue prediction spreadsheets into a ML solution by curating a 250K-row proprietary dataset, optimizing feature selection and engineering, reducing manual data gathering time by 85%.
- Deployed a grid search-tuned Support Vector Regressor, achieving 92% test accuracy (40% improvement over previous methods), driving valuation processes in production for private company market assessments.
- Designed a Keyword-to-Comment extraction NLP tool for financial documents, using Capital IQ API, and Named Entity Recognition, automating data workflows and expediting decision making for market research analysts.
- Uncovered $2.5M in unbilled revenue by developing a MySQL-integrated Tableau dashboard that detected billing anomalies for immediate recovery.
- Engineered a scalable, cloud-native data pipeline using AWS Kinesis, Lambda, and S3, integrating over 30 disparate financial data sources, reducing data latency by 95%.

TALWART, Mohali, India
Digital Marketing Data Analyst — May 2020 – Aug 2022

- Spearheaded data-driven quarterly campaigns, leveraging data segmentation and targeting techniques on customer demographics and purchasing behavior, resulting in 120+ new clients and $350,000 in revenue.
- Utilized Python and MySQL for exploratory data analysis on customer behavior and ad performance datasets, driving tailored A/B Tests on a $110,000 Google Ads budget, resulting in a 28% increase in Return on Ad Spend.
- Developed an anomaly detection tool using MySQL and Python, harnessing Google Analytics data, pinpointing web traffic irregularities, preventing marketing setbacks, and conserving roughly $80,000 in revenue.
- Employed Tableau's predictive modeling and clustering for scenario analyses on ad spend patterns, visualizing potential ROI trajectories, enabling optimized budget allocations, and achieving an 18% reduction in wasteful expenditures.
- Conducted Multivariate Testing in Google Optimize, improving the conversion rate for corporate clients by 35%.

HONEST & YOUNG, Melbourne, Australia
Tax Analyst — Jul 2019 – Mar 2020

- Guided clients through healthcare policy reductions, pre/post-tax deductions, and tax offsets, minimizing their tax liability, while fostering quality client relations resulting in 50+ new clients through referrals.
- Engineered a descriptive analytics model using Oracle DB, VBA, and Excel, decoding intricate tax data patterns and transactional behaviors of rideshare industry clients, streamlining the pre-tax analysis process for 30% of the clientele.
- Reduced client overpayments by 15% by designing a tax optimization tool to dissect multifaceted transactional datasets.

ACADEMIC PROJECTS

Deepfake Detection System *(Stack: SWIN-Transformer, GAN, React UI, Flask)* — Sep 2024 – Dec 2024

- Launched a real-time deepfake detection system, enriching FaceForensics++ dataset with GAN-generated deepfakes, fine-tuning a SWIN-Transformer, predicting real vs fake images with a 96% classification accuracy (F1:97.2%).

Automated Podcast Summarization and Highlight Generation *(Stack: NLTK, Cloud API's)* — Mar 2024 – May 2024

- Revolutionized content synthesis, utilizing LLM's (BART, GPT) alongside with Facebook's Wav2Vec2 speech processing transformer to meticulously transcribe audio, and generate summaries and key highlights (topic-modelling) for 100K+ podcast episodes, elevating user discovery and engagement.

Optimizing Urban Mobility with Advanced Clustering *(Stack: Geopandas, Scikit-learn)* — Feb 2024 – May 2024

- Employed Python, Folio, and clustering (HDBSCAN) to analyze 1.3M+ bike-sharing trips, fusing geospatial, traffic, incident-report, and weather data, recalibrating service deployment strategies for spotlighting safety and optimizing spatial coverage.

Sentiment-Driven Crypto Market Forecasting *(Stack: TensorFlow, transformers, statsmodels, NLTK)* — Oct 2023 – Jan 2024

- Pulled financial data from Coinbase and 250K+ tweets and 3M+ comments via Twitter API, leveraging the BERT LLM for sentiment analysis, integrating sentiment into financial data, and forecasting crypto prices using time series (ARIMA, Prophet) and random forest regressor, achieving 90% accuracy and 20% uptick in ROI.